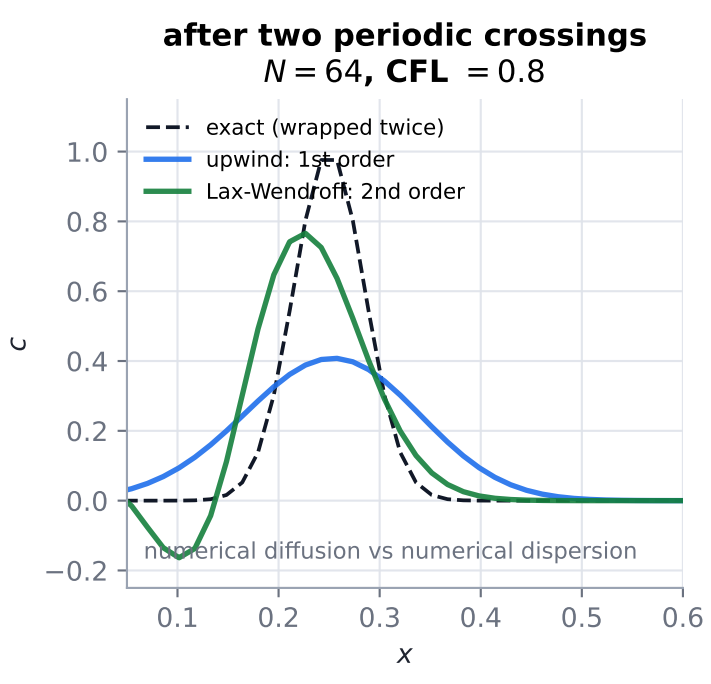
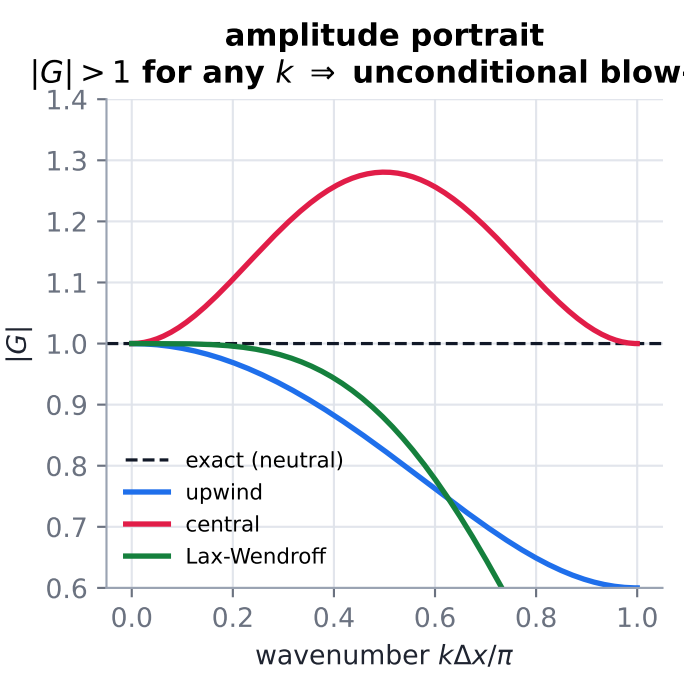

after two periodic crossings
$N = 64$, CFL $= 0.8$
exact (wrapped twice)
upwind: 1st order
Lax-Wendroff: 2nd order
numerical diffusion vs numerical dispersion
c
x


amplitude portrait
$|G| > 1$ for any k ⇒ unconditional blow-up
exact (neutral)
upwind
central
Lax-Wendroff
$|G|$
wavenumber $k\Delta x/\pi$

phase portrait
$c_{up} = c$ ⇒ every wave travels at the wrong speed
exact $\omega = kU$
upwind
Lax-Wendroff
phase advance per step
$k\Delta x/\pi$